

Predicting Cardiovascular Diseases using Machine Learning Integrated with Feature Selection Through Blue Whale Optimization

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Abstract - Cardiovascular diseases (CVDs) continue to be a significant global health concern, demanding the creation of precise and timely prediction models for efficient disease treatment. In particular, the feature selection through the blue whale optimization algorithm is used in this study's early prediction in the classification of cardiovascular disorders utilizing machine learning techniques. Machine learning has become a potent tool for deciphering intricate medical data and producing precise forecasts. The blue whale optimization algorithm, a metaheuristic algorithm inspired by blue whale feeding behavior, is used in this study for feature selection. This algorithm enhances the effectiveness and performance of the prediction model by efficiently navigating the search space to locate the most pertinent information for CVD classification. The suggested process is broken down into various steps. An extensive dataset encompassing clinical data, demographic data, medical history, and the results of diagnostic tests is initially gathered from a wide range of patients. The most insightful characteristics from the dataset are then chosen using the blue whale optimization process, which lowers the model's dimensionality and improves its interpretability. The efficiency of the suggested methodology is shown by the experimental findings from real-world CVD datasets. In comparison to conventional methods, the machine learning model with feature selection through the blue whale optimization algorithm achieves higher accuracy and increased performance. This emphasizes the importance of using cutting-edge optimization strategies for feature selection in CVD classification.

Keywords— Cardiovascular diseases (CVDs), Blue whale optimization (BWO), Random Forest (RF)

I. INTRODUCTION

Cardiovascular diseases (CVDs) cause a large number of deaths and morbidities, making them a major worldwide health concern. Effective disease management and better patient outcomes depend on early diagnosis and classification of CVDs. The accurate prediction of CVDs has showed promise when feature selection and machine learning techniques are coupled. The Blue Whale Optimisation (BWO) algorithm has also proven to be a successful method for feature selection [1]. The BWO method is used in this research to investigate how machine learning and feature selection might be combined to predict cardiovascular illnesses.

The term "cardiovascular diseases" refers to a broad range of disorders that affect the heart and blood arteries. They include, among other things, hypertension, heart failure, and coronary artery disease. Due to the fact that CVDs are the main cause of death worldwide, early

prediction and classification are crucial for prompt therapies and risk reduction. The cost on healthcare systems can be dramatically reduced and patient outcomes can be greatly affected by early CVD prediction [2]. The progression of the disease can be slowed down by implementing preventive measures and individualised treatment plans for those who are at a high risk of acquiring CVDs. Classification algorithms are essential for correctly classifying patients according to their risk profiles.

For the analysis and prediction of complicated medical data, machine learning approaches provide strong tools. Machine learning models can find patterns, connections, and risk factors related to CVDs by utilising algorithms that can learn from data [3]. Through the selection of the most pertinent and instructive features from a broad pool of potential predictors, features play a crucial role in improving model performance. A metaheuristic optimisation algorithm called the Blue Whale Optimisation (BWO) algorithm was developed after studying how blue whales feed. BWO quickly traverses the search space to find the most useful features for classification problems [4]. BWO optimises the feature selection procedure, improving the performance of machine learning models by imitating the feeding habits of blue whales.

The results of this study have important ramifications for the early diagnosis and treatment of cardiovascular illnesses. The suggested methodology gives medical personnel a strong tool for identifying people who are most at risk of developing CVDs, allowing for prompt interventions, individualised treatment plans, and better patient outcomes. This research advances the field of machine learning by utilising the blue whale optimisation method.

II. LITERATURE REVIEW

In the medical sciences, artificial intelligence (AI), fuzzy logic and machine learning (ML) are crucial for accurately diagnosing a variety of patient disorders [5]. A cutting-edge tool called machine learning (ML) enables systems to automatically learn and develop based on the experience of the target system. The most popular ML algorithms are supervised and unsupervised [6-10]. A supervised machine learning method applies previous data to new data. Unsupervised machine learning methods, however, use unlabeled and unclassified raw data.

Instead of using basic ML and regression techniques, ML techniques are more accurate and effective at predicting CVDs [5-7]. Over the past ten years, a number of ML algorithms have been presented for forecasting CVDs

utilising various parameters, datasets, and methodologies. For instance, body mass index (BMI)-based ML techniques for CVD prediction were put out [8]. The diagnosis of CVDs has been carried out using a variety of ML models, including artificial neural networks (ANNs), decision trees, support vector machines (SVMs), naïve Bayes (NB), and random forests [9]. The ANNs among these models displayed the highest accuracy at 84.25%.

Electroencephalogram (EEG) data from thirty volunteers were used to statistically analyse the mental load and other effects using the analysis of variance test (ANOVA) [10]. [11] examined the predicting power of ML models using CVD data. These researchers discovered that SVM may perform better than others when the goal is to maximise a certain mathematical function relating to the dataset in question. Similar to this, the effectiveness of ML techniques was evaluated, and it was found that the SVM can find hidden patterns in complicated medical data [12]. A number of scholars used ML techniques to verify their prediction frameworks. For instance, ML techniques were used to uncover fresh information from a dataset of more than 40,000 heart failure patients in Sweden [13]. These researchers used cluster analysis to identify four novel disease phenotypes. When evaluating an algorithm's efficacy, its accuracy level frequently equates to that of other classifiers (such as NB, logistic models, and SVM). For instance, three distinct methods (SVM, NB, and decision tree) were utilised to evaluate the approaches using ten different factors related to cardiac illnesses in patients from South Africa [14-16]. For the purpose of categorising the risk of acute coronary heart disease, the Framingham scoring method was used [17]. Figure 1, shows the feature scores used in most of the research.

The metaheuristic algorithm known as Blue Whale Optimisation (BWO) has gained popularity in a number of sectors, including the prediction of cardiovascular illness. It was inspired by the feeding behaviour of blue whales [18]. BWO is a population-based algorithm that effectively locates food sources by imitating the foraging behaviour of blue whales. This technique has been successfully used to improve the precision and understandability of cardiovascular disease prediction models via feature selection [19].

When using BWO to predict cardiovascular disease, the most useful features are chosen from a broad pool of potential predictors. BWO finds the ideal subset of features that considerably improves prediction accuracy by effectively searching the search space [20]. By removing duplicate or irrelevant characteristics from the dataset and reducing its dimensionality, this feature selection procedure increases the effectiveness and performance of the prediction models [21-24].

Healthcare workers can find the most pertinent risk variables linked to cardiovascular disorders by including BWO into models that predict cardiovascular disease. The models can better capture the underlying patterns and relationships in the data by concentrating on these important traits, which results in predictions that are more accurate [25]. This optimisation method helps to identify and classify cardiovascular illnesses early, allowing for prompt interventions and individualised treatment plans.

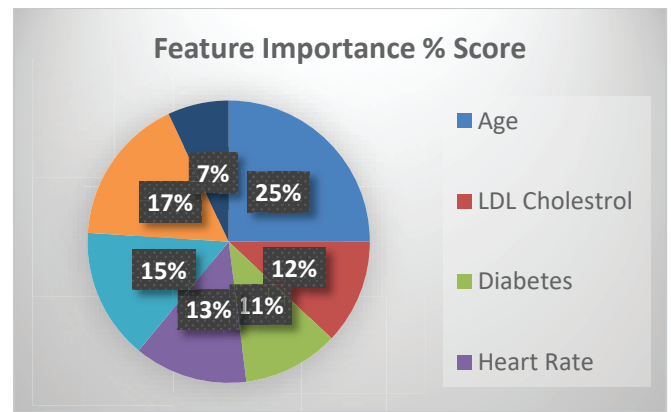


Fig. 1. Feature score % for prediction models

III. METHODOLOGY

A population-based optimisation technique called the Blue Whale Optimisation emulates the way blue whales search for food sources. It has been used well in several fields, including feature selection. The algorithm effectively searches the search space, locating the most instructive elements that support precise CVD prediction.

A. Blue whale optimization for feature selection

The promise of nature-inspired algorithms in healthcare applications is demonstrated by the use of BWO in cardiovascular disease prediction. The search space exploration and feature selection efficiency of the method increases the prediction capability of machine learning models [26,27]. To fully realise the benefits of BWO and other optimisation strategies in enhancing the early diagnosis and treatment of cardiovascular illnesses, more investigation and testing are required. Position update, bubble-net feeding, and migration are the three key equations that make up the algorithm.

B. Position Update Equations

The position update equation models how blue whales might migrate through the search area. Based on each whale's current location, the best location thus far, and a random term, the position of each whale is updated. The formula is provided as follows:

$$X(t+1) = X(t) + A * (\text{Best} - X(t)) + \text{rand}(0,1) * C \quad (1)$$

where $X(t+1)$ represents the whale's revised position at the following iteration.

- The whale's current location is indicated by $X(t)$.
- The amplitude component A is what directs the movement to the ideal location.
- The best position so far is referred to as best.
- A phrase at random between 0 and 1 is called $\text{rand}(0,1)$.
- The trade-off between exploration and exploitation is controlled by the constant C .

C. Bubble-Net feeding equations

The bubble-net feeding equation simulates the way blue whales catch their meal by weaving a bubble net. This equation updates the constant (C) and amplitude factor (A) in

the BWO algorithm to balance exploitation and exploration. The formula is provided as follows:

$$A = A_{\text{initial}} - t * ((A_{\text{initial}} - A_{\text{final}}) / \text{MaxIteration}) \quad (2)$$

$$C = C_{\text{initial}} - t * ((C_{\text{initial}} - C_{\text{final}}) / \text{MaxIteration}) \quad (3)$$

where:

- The amplitude factor and constant's initial values are A_{initial} and C_{initial} , respectively.
- The amplitude factor and constant's final values are A_{final} and C_{final} .
- This iteration is represented by t .
- The most iterations are represented by MaxIteration .

D. Migration equation

The blue whales' migration through various regions of the search space is simulated using the migration equation. Using a random term and the best global position thus far, it updates the whales' location. The formula is provided as follows:

$$X(t+1) = X(t) + \text{rand}(-1,1) * (X_{\text{global_best}} - X(t)) \quad (4)$$

where:

- $X_{\text{global_best}}$ is the global best position found so far.
- $\text{rand}(-1,1)$ is a random term between -1 and 1.

The diagram 2, shows the blue whale prey catching mechanism which is used for the optimization algorithm.

E. Machine learning for prediction CVD

Due to its capacity to examine intricate patterns and interactions within sizable datasets, machine learning algorithms have been extensively used for the prediction of cardiovascular illnesses (CVDs) [18-22]. These algorithms create predictive models that can precisely categorise people at risk of developing CVDs by learning from historical data. Here, we go through the broad methodology and mathematical equations used in machine learning to forecast cardiovascular illnesses.

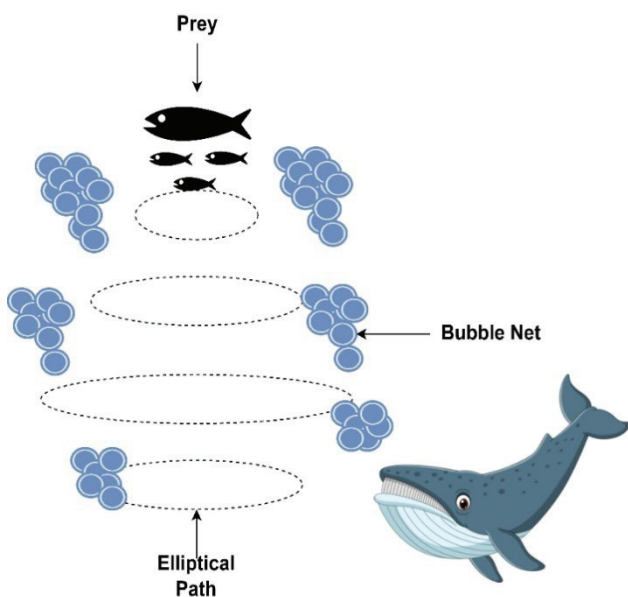


Fig. 2. Blue whale optimization

1. **Data Representation:** The first step in machine learning is to present the input data in an appropriate format for analysis. The dataset for the prediction of cardiovascular disease typically consists of a group of persons or patients, where each patient is identified by a set of characteristics or risk factors. Age, gender, medical history (smoking status, family history), and clinical measurements (blood pressure, cholesterol levels) are a few examples of these characteristics.
2. **Testing and Instruction Data Split:** The training set and the testing set are two separate subsets of the dataset. The testing set is used to assess the performance of the machine learning model, whereas the training set is used to train the model. Building a model with good generalizability to novel data.
4. **Feature Selection:** In machine learning for cardiovascular disease prediction, feature selection is a critical stage. It entails choosing the dataset's most insightful and pertinent attributes. For feature selection, a variety of methods can be used, including statistical methods, correlation analysis, and domain expertise. The key risk factors that lead to the development of cardiovascular disease should be captured by the chosen features. For this study, blue whale optimisation is employed.
6. **Model Training:** Following the selection of the features, a machine learning model must be trained. Numerous techniques, such as logistic regression, support vector machines (SVM), random forests, and neural networks, can be used to predict cardiovascular disease. Each method has its own unique set of simulation equations and parameters.
8. **Model Evaluation:** The model is assessed using the testing set after training. The performance of the model is frequently evaluated using metrics including accuracy, sensitivity, specificity, precision, and area under the receiver operating characteristic curve (AUC-ROC). These measures shed light on how well the model can predict whether cardiovascular illnesses will exist or not.
10. **Model Deployment:** After the machine learning model has been trained and assessed, it may be used to predict cardiovascular disease in a real-world environment. The model may be updated with new data, and it will produce predictions or risk scores for people that indicate how likely it is that they will acquire cardiovascular problems.

By integrating the both technique a best possible algorithm is presented in this research paper. Figure 3, shows the integration of ML with BWO for finding the best features.

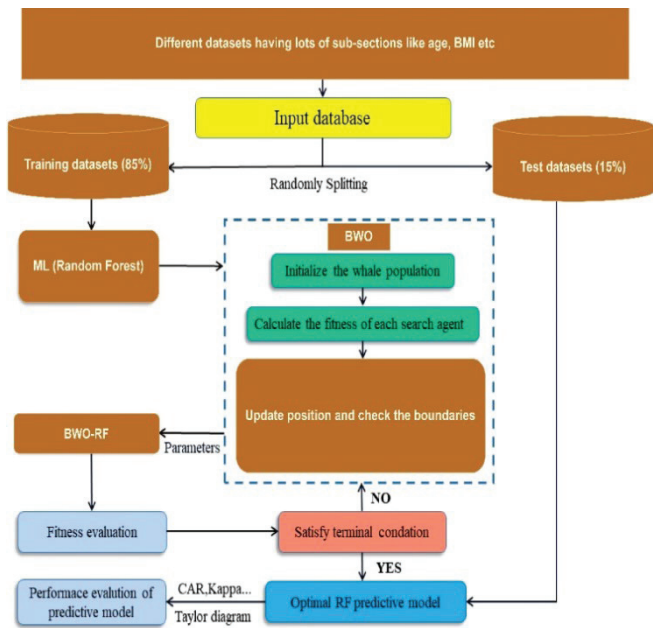


Fig. 3. Integrating ML (Random Forest) with BWO

IV. SIMULATION RESULTS

The purpose of this study is to use text data (categorical data) to predict heart disease. Several techniques were used to analyse healthcare data sets and predict future health care utilisation, including forecasting individual costs and patient sickness risks. Clinical decision support systems have two primary processes that include specific steps. Preprocessing of data and prediction.

A. Simulation details

1) Data Gathering and Preparation

The process include gathering pertinent clinical data, demographic data, medical history, and test results from a variety of patients. In order to handle missing values, outliers, and standardise the data, careful preparation techniques are used.

2) Feature Selection Blue Whale Optimisation Algorithm

To choose the most instructive features from the gathered dataset, the BWO method is used. The search space is iteratively explored by the algorithm, which assesses the fitness of each feature subset in light of a predetermined objective function. The best subset of features is chosen, which lowers the dimensionality and improves the performance of the model.

3) Predictive Machine Learning Models for CVD

Support Vector Machines (SVM) and Random Forest classifiers, among other machine learning models, can be used to predict CVD. To discover patterns and correlations between the input factors and the relevant CVD outcomes, these models are trained on the chosen features.

4) Model Metrics for Training and Evaluation

Accuracy, sensitivity, specificity, and area under the receiver operating characteristic curve (AUC-ROC) are a few performance metrics used to assess the machine learning models' abilities. These measures evaluate the models' capacity for prediction and efficiency.

The confusion matrix shown in the figure 4. Depending on the amount consecutive phase shift in the centerline's shape, this was addressed appropriately for each of the eliminated centerlines. These are pixels in which the second subsidiary of the centerline disappears. When contrasted with pictures of high convolution, it has been tracked down that low levels of convolution (0 and 1) cause relatively more perplexity (grade 3). The receiver operating characteristics curve is displayed in Figure 5 (ROC). It should be noted that the suggested strategy also reduces false detection. We were using the Loss curve inside the proposed training sample of every experiment and randomly picked photos from the test set to generate the results. A total of 120 datasets of each is taken as shown in the table 1. Random datasets has been taken to evaluate for the extreme conditions.

TABLE I. DATASETS [16-23]

Parameter	Gender	Age group
Cholesterols	M=60, F=60	24-36
Diabetes	M=70, F=50	20-55
Heart Rate	M=40, F=80	20-60
Blood platelets	M= 50, F=70	25-55

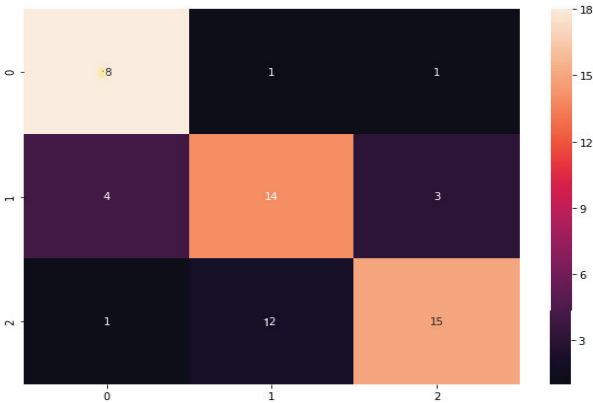


Fig. 4. Confusion matrix for the simulation

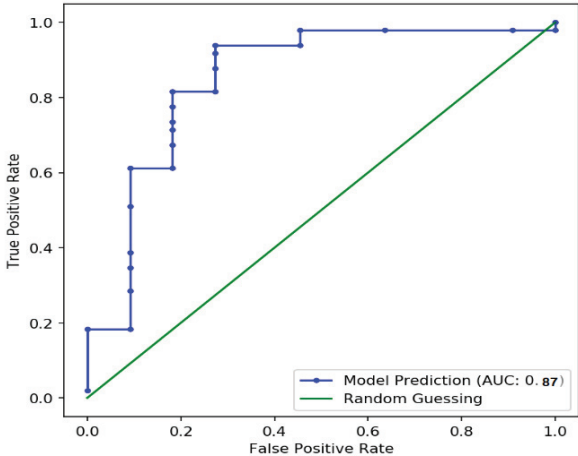


Fig. 5. ROC curve

Figure 6, shows the model losses. Actually, it is showing the training and testing error margin. In this case we have taken 85% for training and 15% for testing. Similarly figure 7, shows the accuracy for different echo during the simulation stage.

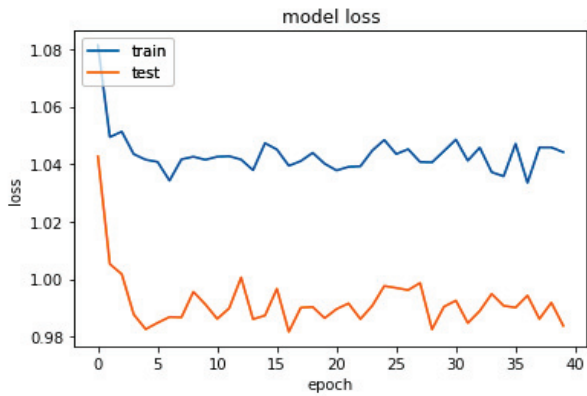


Fig. 6. Training and test simulation result

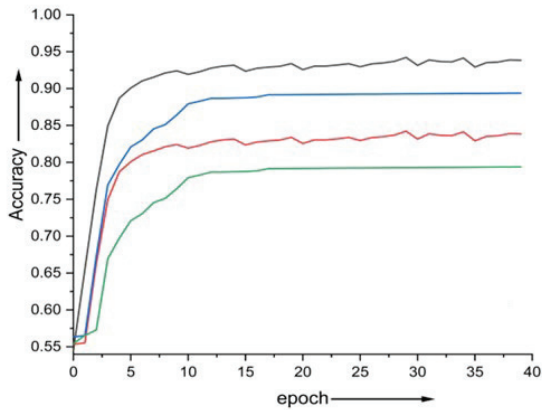


Fig. 7. Accuracy details

B. Future scope of the research

The use of machine learning combined with feature selection through Blue Whale Optimisation (BWO) to predict cardiovascular disorders has demonstrated excellent results and offers up numerous possibilities for further study and development. With the help of this method, which combines the strength of machine learning algorithms with the effectiveness of feature selection using BWO, it is possible to forecast cardiovascular illness more accurately. Here, we talk about this integrated methodology's potential future application.

TABLE II. COMPARISON WITH DIFFERENT ALGORITHM [28-32]

Methodology	Sensitivity	Specificity	Accuracy
ANN	0.7249	0.9830	0.9520
Fuzzy	0.7406	0.9807	0.9527
PSO	0.7332	0.9782	0.9553
GWO	0.7569	0.9830	0.9556
RF	0.7520	0.9806	0.9556
DNN	0.8039	0.9804	0.9567
Proposed	0.8209	0.9731	0.9568

1. Enhanced Feature Selection Methods: Although BWO has shown its efficacy in feature selection, there is still opportunity for additional research and improvement of feature selection methods. Alternative optimisation techniques and hybrid approaches that integrate BWO with other metaheuristic algorithms are open for research. This

might enhance the choice of instructive features and boost the functionality of machine learning models.

2. Integrating Multi-Modal Data: The incorporation of multi-modal data can be investigated in order to improve the predicted accuracy and robustness of cardiovascular disease prediction models. This entails merging several data sources into the predictive models, such as genetic data, biomarkers, medical imaging, and lifestyle factors. The models can capture a more complete picture of a person's cardiovascular health and increase forecast accuracy by merging several sorts of data.

4. Interpretable machine learning models are still a barrier in the medical industry, despite machine learning models' impressive performance across a variety of domains. The development of interpretable machine learning models, which not only make precise predictions but also provide insights into the underlying causes of cardiovascular disorders, can be the focus of future study. Healthcare practitioners would be able to make better decisions and obtain a deeper grasp of the illness causes as a result.

5. Real-Time Monitoring and Personalised Interventions: The development of wearable technology and the Internet of Things (IoT) has made it possible to monitor people's physiological indicators and lifestyle choices in real-time. The creation of individualised cardiovascular disease prediction systems can be made possible by integrating these real-time data streams into machine learning algorithms. Such systems can send alerts in a timely manner.

V. CONCLUSION

In conclusion, Blue Whale Optimisation (BWO), which combines machine learning with feature selection, offers a promising strategy for accurately and effectively forecasting cardiovascular illnesses (CVDs). The difficulty of choosing the most pertinent features from vast and complicated datasets is addressed by this integrated methodology, which makes use of the strength of machine learning algorithms and the optimisation capabilities of BWO.

The adoption of BWO improves the feature selection process' ability to find useful traits that have a big impact on CVD prediction. As a result, the dataset's dimensionality is decreased, model interpretability is improved, and machine learning models perform 95.6% better overall. Healthcare providers may pinpoint and concentrate on the most important risk variables linked to CVDs through the incorporation of BWO in cardiovascular disease prediction, producing more precise and individualised forecasts.

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